Aumazing: A Gamified Learning App for Children with Early Childhood Autism Spectrum Disorder

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CHAPTER I

INTRODUCTION

1.1 BACKGROUND OF THE STUDY

The rapid computerization of education and healthcare has transformed early childhood intervention through the use of mobile applications, artificial intelligence, and gamified technologies (Justicia et al., 2026b; Xu et al., 2025). These digital tools are increasingly used to provide adaptive support, automate performance analysis, and personalize learning experiences. For children with Autism Spectrum Disorder (ASD), technology-mediated interventions offer accessible, visual, and reward-based activities that can supplement traditional therapy and support developmental learning at home (Justicia et al., 2026b; Micai et al., 2024).

Globally, Autism Spectrum Disorder (ASD) is recognized as a significant developmental and public health concern, with the World Health Organization estimating that autism affects approximately 1 in 100 children and emphasizing the need for early support and inclusive services (World Health Organization, 2025). Technology-mediated interventions have emerged as practical support mechanisms, and a systematic review found that telemedicine-delivered interventions for autistic children and adolescents were comparable in effectiveness to in-person approaches in several contexts and were often associated with lower costs (Micai et al., 2024). In the Philippines, affordability remains a major challenge, as a local study reported that the mean post-diagnosis intervention cost in the first year was ₱38,868, or around ₱3,239 per month, creating a substantial burden for many households; the same study also noted that national prevalence data on autism remain limited, making service planning more difficult (Quilendrino et al., 2022). This concern is further supported by UNICEF Philippines, which highlighted that families raising children with disabilities often experience higher costs and greater economic strain (UNICEF Philippines, 2022). At the local level, Davao City has strengthened developmental support through the Davao City Special Needs Intervention Center for Children (DCSNICC), which has served 536 children with special needs since its soft opening and provides occupational therapy, physical therapy, speech and language therapy, and early intervention services, showing both strong demand and the continuing need for tools that can reinforce therapy at home (City Government of Davao, 2025).

In this context, the proposed capstone project, Aumazing: A Gamified 2D Learning Application with AI-Driven Assessment for Children with Early-Stage Autism Spectrum Disorder, will be conducted to provide a supplemental system for early

intervention and home-based therapy. The proposed system will include gamified pre-assessments, leveled learning modules, post-assessments, AI-based performance analysis, a parent dashboard, screen time management, and a premium therapy gateway with video conferencing and therapy center directory features. At present, many parents, especially mothers as primary caregivers, face difficulty in consistently guiding children with early-stage autism through home-based activities due to time constraints, lack of training, and limited access to specialists. These challenges slow developmental progress and create gaps between clinical sessions and home practice. Through the development of this system, the proponents aim to improve the current process of home-based support, provide structured and engaging intervention activities, and help address the practical problems encountered by families in sustaining autism therapy outside formal clinical settings (World Health Organization, 2025; Quilendrino et al., 2022; Micai et al., 2024; City Government of Davao, 2025).

1.2 OBJECTIVES OF THE STUDY

The general objective of the study is to develop a gamified 2D mobile learning application that utilizes AI-driven pre-assessment and analytics to support the early intervention and home-based therapy of children aged two to six with early-stage Autism Spectrum Disorder.

Specifically, this study aims:

- To design and develop a 2D gaming environment that incorporates gamified pre-assessments, leveled learning modules, and post-assessments tailored to the behavioral and psychological needs of children with an early-stage ASD.
- To implement an AI-based assessment framework using XGBoost to analyze child performance during gameplay and a rule-based with content-based recommendation approach to recommend specific learning modules and track developmental progress.
- To incorporate a screen time management system and a parent dashboard to ensure balanced usage and provide mothers with data-driven insights into their child's improvement.
- To integrate a premium therapy gateway featuring video conferencing and a directory of therapy centers to facilitate professional-led remote sessions and physical consultations.

1.3 SIGNIFICANCE OF THE STUDY

This study will be beneficial to several groups.

Children with early-stage ASD. The study will provide them with an engaging and structured 2D learning environment that supports the development of basic cognitive and behavioral skills through interactive and visually guided activities.

Parents, especially mothers. The application will help them monitor their child's progress, manage screen time, and identify appropriate learning modules based on assessment results. It can also lessen the difficulty of planning home-based intervention activities.

Therapists and therapy centers. The system will offer an additional platform for reaching clients through remote sessions and directory-based referrals. It may also serve as a supplementary tool for monitoring developmental patterns outside formal therapy sessions.

Schools and academic institutions. The study may contribute to the development of inclusive educational technologies and strengthen research on gamification, mobile learning, and AI-assisted support systems for children with developmental needs.

Future researchers. The findings and system design of this study may serve as a basis for future improvements, additional modules, broader autism-support tools, and more advanced AI-based intervention systems.

1.4 SCOPE AND DELIMITATION

This outlines the comprehensive scope of the research, particularly concerning the development and initial deployment of the digital platform, while also clearly defining the deliberate delimitations necessary for its focused and effective execution.

1.4.1 Scope

The study covers the design and development of **Aumazing**, a gamified 2D mobile learning application intended to support the early intervention and home-based

therapy of children aged **two to six years old** with **early-stage Autism Spectrum Disorder (ASD)**. Specifically, the system will include the following features and functions:

- **Mobile-based application deployment** for Android and/or iOS devices to provide accessible home-based usage for parents and children.
- **User registration and login** for parents or guardians to securely access the application and manage child-related records.
- **Child profile management** that stores basic child information such as name, age, developmental details, and user settings relevant to gameplay and assessment.
- **Gamified pre-assessment module** that evaluates the child's initial performance before entering the recommended learning activities.
- **AI-driven assessment analytics** using **XGBoost** to examine gameplay indicators such as response time, answer accuracy, errors, retries, time spent, completed activities, completion rate, and interaction patterns in order to determine the child's initial level.
- **2D interactive learning games** designed to develop basic cognitive and behavioral skills through child-friendly, visually guided, and task-based activities.
- **Leveled learning modules** that organize activities according to difficulty and developmental appropriateness.
- A **module recommendation system** using **rule-based logic with content-based recommendation** to assign the proper starting level or succeeding activities for the child based on assessment results and identified developmental needs.
- **Post-assessment feature** that measures the child's performance after completing a level or module.
- **Progress tracking system** that records and monitors:
    - completed activities,
    - level progression,
    - assessment results,
    - and comparative improvement from pre-assessment to post-assessment.

- **Parent dashboard** that presents the child's assessment summaries, progress reports, improvement metrics, and recommended modules in an understandable format.
- **Screen time management feature** that helps parents monitor and regulate the child's usage duration within the application.
- **Reward- or progression-based game structure** to encourage motivation, repetition, and continued engagement during learning activities.
- **Premium therapy gateway** that provides additional services for users who avail of premium access.
- **Video conferencing feature** under the premium module for possible remote consultation or guided sessions with professionals.
- **Therapy center directory or access feature** that lists available therapy-related centers or practitioners for consultation and support purposes.
- **Local data storage and account-based data retrieval** to preserve assessment records, progress history, and user information.
- **System testing and evaluation** involving selected parents and qualified validators such as psychometricians, SNeD or SpEd teachers, or practitioners to assess the relevance, usability, and appropriateness of the application.

1.4.2 Delimitation

The study is subject to the following limitations and exclusions:

- The application is **not intended to diagnose Autism Spectrum Disorder** and shall not replace a formal clinical diagnosis by a pediatric neurologist, developmental pediatrician, psychologist, or other qualified medical specialist.
- The system is only intended as a **supplementary support, screening, and home-based intervention tool**.
- The application will only focus on **children aged two to six years old** with **early-stage ASD** and is not designed for older age groups or for other developmental conditions outside the defined target coverage.

- The application will only include **selected basic cognitive and behavioral learning activities** and will not cover the full range of clinical, academic, speech, occupational, or behavioral therapy programs.
- The AI component will only function for **assessment analytics and module recommendation** based on gameplay performance; it will not serve as an autonomous diagnostic, therapeutic, or medical decision-making system
- The recommendation output of the system will be limited to **in-app learning modules** and will not generate formal medical, psychiatric, or psychological findings.
- The premium therapy gateway will only provide **video call access and therapy center directory features**; it will not guarantee therapist availability, treatment outcomes, or official therapy center partnerships unless formally arranged by the proponents.
- The system will not include **physical therapy execution**, face-to-face intervention, or direct therapist-administered treatment inside the application.
- The application will be limited to **mobile devices only** and will not initially support desktop or web-based full system deployment.
- Some system functions, particularly:
    - AI assessment syncing,
    - cloud-based data updates, and video conferencing, will require a **stable internet connection**.
- Offline use may only apply to **selected 2D game activities**, and full synchronization of analytics and premium services may not be available without internet access.
- The study will not include **large-scale clinical trials** or long-term medical outcome validation.
- The content and recommendations of the app will be based only on the **designed modules, defined assessment indicators, and validator feedback** gathered during the scope of the capstone project.
- The project is limited by the **time frame, resources, and technical capacity** of the proponents from development up to testing and documentation.

1.5 TIME AND PLACE OF THE STUDY

The development and evaluation of the proposed system will be conducted from February 2026 to September 2026, encompassing the planning, design, development, testing, deployment, and documentation of the project. System development and documentation will be undertaken at the proponents' institution, while consultation, validation, and possible data gathering may be conducted with selected parents, psychometricians, SNeD or SpEd teachers, and other qualified practitioners within Davao City or nearby accessible areas to ensure the application's relevance and effectiveness.

1.6 OPERATIONAL DEFINITION OF TERMS

This section defines key technical terms used in the Aumazing: A Gamified 2D Learning Application with AI-Driven Assessment for Children with Early-Stage Autism Spectrum Disorder capstone project to ensure clarity and understanding.

1. **Artificial Intelligence (AI).** In this study, AI refers to the hybrid assessment and recommendation framework of the system, where **XGBoost** is used to analyze the child's gameplay performance and **rule-based with content-based recommendation** is used to suggest suitable learning modules.
2. **Autism Spectrum Disorder (ASD).** A developmental condition affecting communication, social interaction, behavior, and learning patterns. In this study, the term refers specifically to children aged two to six who are in the early stage of the condition.
3. **Gamification.** The application of game elements such as levels, rewards, challenges, and interactive tasks to increase motivation and participation in learning activities.
4. **Home-based Therapy.** Supportive intervention activities performed at home by parents or guardians to reinforce learning and behavioral development.
5. **Parent Dashboard.** A system feature that displays progress data, insights, and usage information to help parents monitor their child's development.
6. **Post-Assessment.** An evaluation conducted after completion of a module or activity to measure the child's improvement.

7. **Pre-Assessment.** An initial assessment conducted before the learning modules to determine the child's starting skill level and recommend appropriate activities.
8. **Premium Therapy Gateway.** A feature that includes video conferencing and a directory of therapy centers for professional-led remote sessions and consultation access.
9. **Screen Time Management.** A system feature designed to regulate or monitor the child's usage of the application to maintain balanced engagement.

1.7 CONCEPTUAL FRAMEWORK

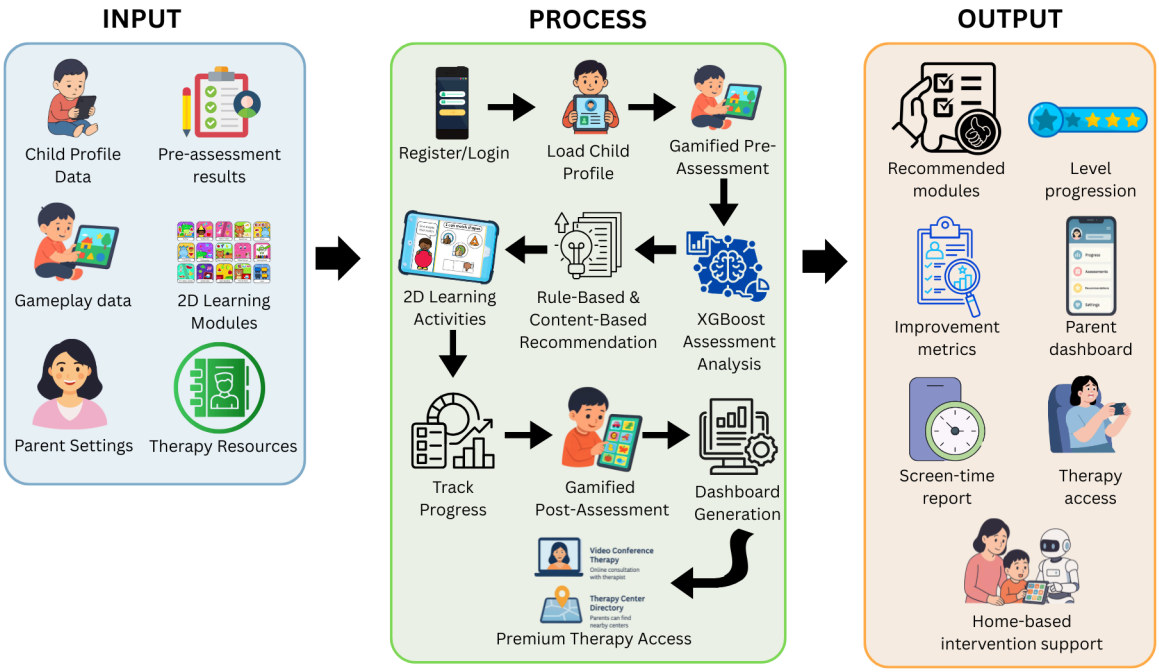

Figure 1: Conceptual Framework

The conceptual framework of the proposed system follows the Input–Process–Output model. In the input phase, the system gathers child profile data, parent information, pre-assessment responses, gameplay performance data, leveled learning content, screen-time settings, and therapy-related resources. In the process phase, the application performs gamified pre-assessment, analyzes child performance using an using XGBoost-based assessment analytics, and applies rule-based and content-based recommendation logic to recommend suitable learning modules, delivers

2D game-based activities, conducts post-assessment, and generates comparative analytics for the parent dashboard while also supporting premium therapy access. In the output phase, the system produces personalized module recommendations, level progression results, post-assessment outcomes, improvement metrics, dashboard reports, and therapy access features that support early intervention and home-based therapy for children with early-stage Autism Spectrum Disorder.

CHAPTER II

REVIEW OF RELATED LITERATURE, STUDIES, AND SYSTEMS

This chapter reviews recent literature, studies, and systems relevant to the AUMAZING capstone project. It focuses on 2D learning games, educational game engagement, mobile learning, and AI-Driven assessment integration. The review highlights recent (last five years) approaches to studies, gamified learning and digital technologies designed to support children with Autism Spectrum Disorder (ASD).

2.1 Autism Spectrum Disorder and Early Intervention

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition that affects behavior, communication, social interaction, and sensory processing (American Psychiatric Association, 2022). Because symptoms vary widely, autism is described as a spectrum. Children with ASD often experience difficulty understanding social cues, communicating verbally, and adapting to environmental changes.

Early childhood is a critical period for intervention because many developmental foundations form between ages two and six. Structured and repetitive activities during this stage can improve cognitive, behavioral, and communication skills. Early intervention programs have been shown to enhance language development, social skills, and adaptive behavior among children with ASD (Zwaigenbaum et al., 2021).

Children with ASD benefit from structured learning environments that include predictable routines, visual guidance, and interactive activities suited to their learning pace. Technology-assisted learning tools have also been found to help develop behavioral and developmental skills more effectively (Odom et al., 2020). Early behavioral intervention further contributes to improved long-term outcomes (Lord et al., 2020).

Home-based support is also important since parents and caregivers spend the most time with the child. However, many parents may lack professional training to implement therapy activities consistently. Digital tools and mobile applications can help guide

parents by providing structured learning activities and progress monitoring (Parsons & Cobb, 2021).

2.2 Behavioral and Psychological Foundations of the Study

The design of the learning modules in this study is guided by several behavioral and psychological approaches for children with ASD. One widely recognized approach is Applied Behavior Analysis (ABA), which uses structured teaching, repetition, and reinforcement to encourage positive behaviors and learning outcomes (Cooper, Heron, & Heward, 2020).

Within ABA, Discrete Trial Training (DTT) breaks complex skills into smaller tasks taught through repeated practice and immediate feedback. This method promotes focused learning and consistent progress among children with ASD (Leaf & McEachin, 2022).

Another theoretical basis is the TEACCH approach, which emphasizes structured environments and visual learning. These strategies reduce confusion and help children focus on tasks more effectively (Mesibov & Shea, 2022). Visual prompts, organized layouts, and predictable gameplay align well with a 2D learning application.

Research shows that structured visual learning environments increase engagement and task completion among children with autism (Hume, Boyd, Hamm, & Kucharczyk, 2021).

2.3 Gamification in Learning Applications

Gamification refers to the integration of game-design elements and principles into non-game contexts with the goal of enhancing engagement, motivation, and participation. In education, gamification utilizes mechanics such as points, rewards, leveled progression, and continuous feedback systems to effectively transform traditional instruction into dynamic, interactive learning experiences (Hussein et al., 2023).

Recent studies consistently demonstrate that gamified learning significantly improves motivation, focus, and task completion rates among learners. For young children, incorporating elements like challenges and rewards makes learning more

immersive and enjoyable, naturally encouraging continued, self-driven participation by tapping into their intrinsic motivation (Ramos Aguiar et al., 2023; Zlotnik et al., 2024).

Gamification is particularly useful for children with Autism Spectrum Disorder (ASD) because it provides structured tasks, necessary repetition, and clear objectives that align with their cognitive profiles. Furthermore, the use of immediate visual feedback, custom learning, and step-by-step progression directly supports the predictable and routine-based learning patterns required to effectively enhance their communication and behavioral skills (Wang et al., 2025; Wythe, 2025).

2.4 AI-Driven Assessment in Learning Systems

Artificial intelligence (AI) is widely used in adaptive learning systems because it can analyze learner performance and adjust content based on individual needs. In this study, XGBoost is proposed as the AI model for assessment because it is suitable for structured gameplay data such as response time, accuracy, errors, retries, time spent, and interaction patterns. Recent studies show that tree-based methods such as XGBoost perform well in classification tasks involving tabular data, making them appropriate for analyzing measurable behavioral indicators in learning systems (Gligorea et al., 2023; Velarde et al., 2024).

For recommendation, the system uses a hybrid approach combining rule-based logic and content-based recommendation. This approach fits the project because the system recommends learning modules based on the child's assessment results and the characteristics of available activities, rather than relying on large user histories. AI recommender systems in education are designed to support personalized learning paths by matching learner needs with appropriate content, while technology-based interventions have also been recognized as effective in providing structured and individualized support for children with Autism Spectrum Disorder (ASD) (Ma, 2025; Oubalahcen et al., 2023; Scarcella et al., 2023). Therefore, combining XGBoost for assessment and a hybrid recommender for module selection allows the proposed system to provide data-informed, personalized, and developmentally appropriate learning support.

2.5 Mobile Applications for Home-Based Therapy

Mobile applications have become effective tools for home-based therapy and learning. Smartphones and tablets provide convenient access to educational resources, allowing children to practice developmental skills outside clinical environments (Alzahrani, 2024; Dubey et al., 2024).

For children with ASD, repeated practice is essential for developing communication, cognitive, and behavioral skills. Mobile platforms allow activities to be repeated consistently while maintaining engagement through visual and interactive elements (García-Moya & Blanco, 2024; Yakubova et al., 2024).

Many applications include animations, sound feedback, and interactive exercises that help sustain attention and reinforce positive learning behaviors through structured gamification (Chapin et al., 2023; Zainuddin et al., 2024).

2.6 Parental Involvement and Monitoring

Parental involvement plays a key role in supporting children with ASD. Since most learning activities occur at home, parents often become primary facilitators of early intervention. Consistent parental support and active involvement in caregiver-mediated interventions have been shown to significantly improve communication, social development, and adaptive skills among children with autism (Wang et al., 2024).

However, parents frequently face severe challenges such as limited time, high parenting stress, and uncertainty about their child's developmental trajectory. This psychological distress is often exacerbated when parents lack the necessary tools, resources, or guidance to cope with the unique behavioral demands of caring for an autistic child, making it difficult for them to identify specific areas requiring further support (Hussain et al., 2023; Pérez-Sola et al., 2024).

Digital learning platforms and telehealth technologies can help address these challenges by providing structured activities paired with continuous monitoring tools. Tracking features allow parents to observe performance patterns and learning progress in real-time. By utilizing digital health interventions equipped with progress dashboards, caregivers are empowered with objective, data-driven insights that not only track the child's cognitive development but also significantly reduce parenting stress and enhance overall caregiving capacity (Jiang et al., 2025; Lu et al., 2024)

2.7 Therapy Center Access and Remote Support Systems

Access to professional therapy services can be challenging for families due to distance, limited specialists, or financial constraints (Zwaigenbaum et al., 2021). These barriers may delay early intervention.

Digital technologies such as telehealth platforms provide remote consultation and monitoring services. Therapists can offer guidance, conduct assessments, and supervise intervention strategies through online communication (Parsons & Cobb, 2021).

2.8 Otsimo | Special Education

Otsimo is a mobile-based, gamified special education platform designed for children with developmental disorders like autism, ADHD, and Down syndrome, focusing on enhancing cognitive, communicative, and academic skills. It utilizes a customized, play-based approach including speech therapy tools and interactive games with positive reinforcement to help children learn and reach their full potential. The platform offers accessible learning, allowing educators and parents to monitor progress in a fun, safe, and engaging environment, often used to bridge the gap between traditional therapy and daily practice. (Otsimo, 2021).

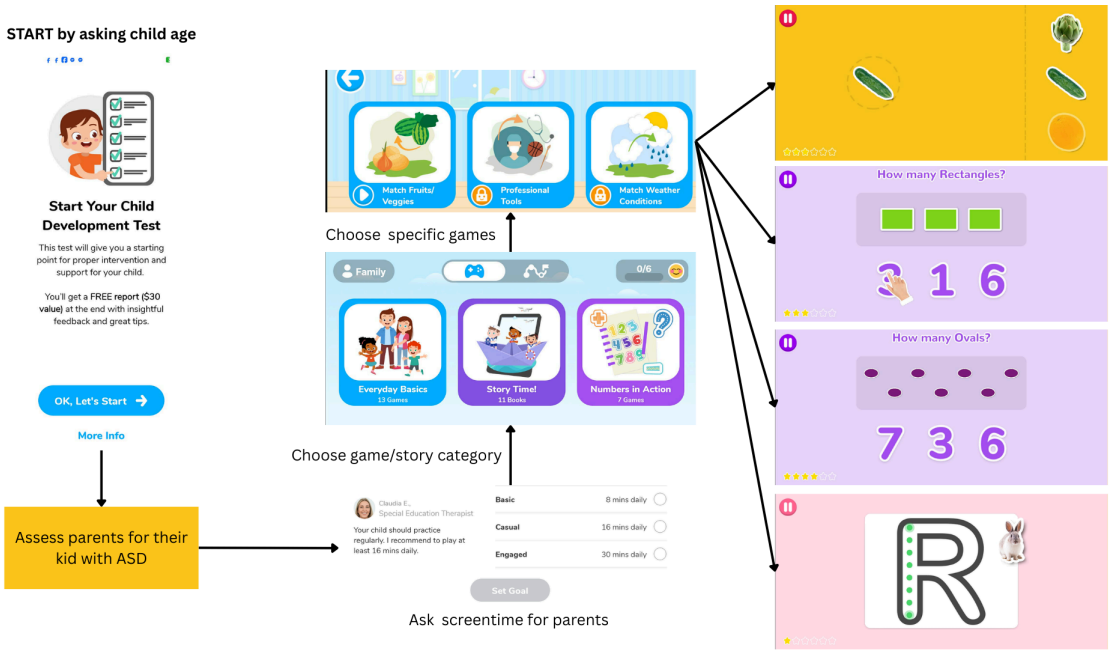

Figure 1: Otsimo mobile app flow for educational games such as matching an object, basic counting shape, letter drawing.

2.9 AutiSpark: Kids Autism Games

AutiSpark is a one-of-a-kind educational app for children with Autism Spectrum Disorder (ASD) with specially designed learning games and approved by experts. If parents are struggling to teach basic concepts to their child, AutiSpark is a must-try for them. AutiSpark offers a multitude of well-researched, engaging and interactive learning games carefully designed to suit the child's learning requirements. Includes concepts of picture association, understanding emotions, recognition of sounds and much more. (AutiSpark, 2022).

Figure 2: Autispark offers many free games and story for kids with autism.

2.10 Proloquo2Go AAC: Tap images. Type Words. Speak.

Proloqu02Go is an easy to use communication app for people who cannot speak or need help being understood. Featuring natural sounding voices, including real children's voices, Proloqu02Go is a simple yet powerful AAC (augmentative and alternative communication) tool. The app is fully customizable and can be utilized to build language skills and grow with a person's communication, supporting both

beginning and advanced communicators. Parents, therapists, teachers and users around the world recognize Proloqu02Go as the gold standard in AAC solutions. The app is by people with autism, Down syndrome, cerebral palsy, Angelman syndrome, and others with difficulties. The app is available in English, Spanish, French, and Dutch and supports bilingual use, so you can switch languages on the fly. (AssistiveWare, 2021).

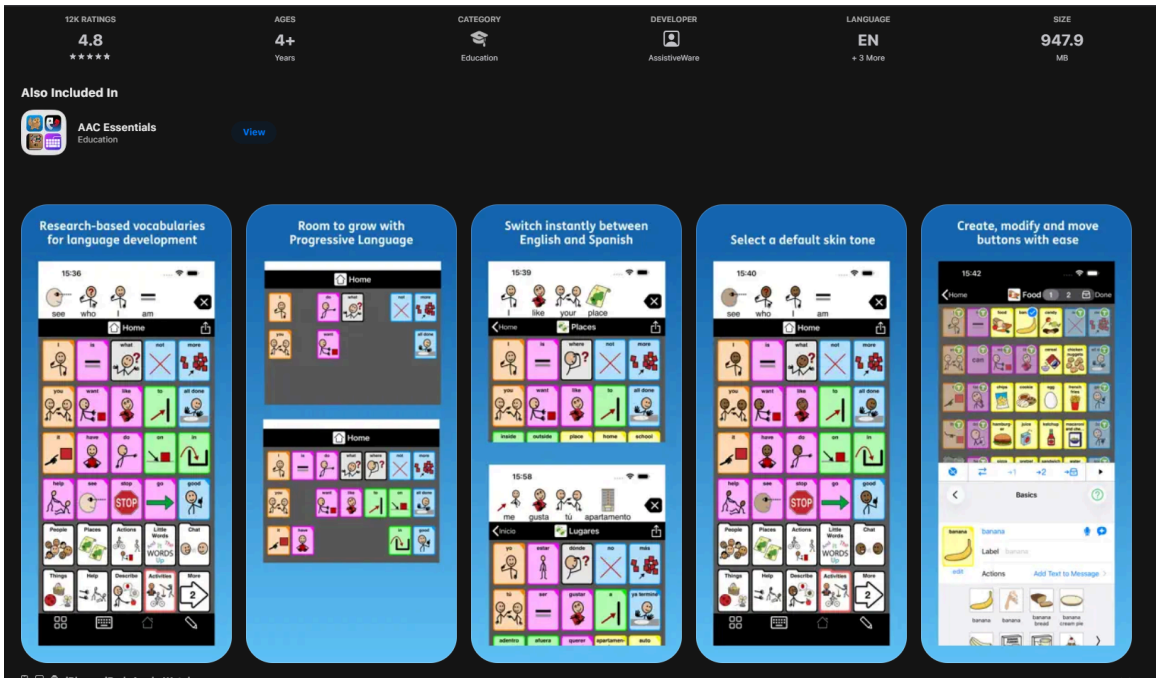

Figure 3: Proloque2Go AAC in App Store

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